

Computer Vision Analysis of Volleyball Serves

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Abstract—Volleyball serve analysis is typically performed through manual video review, which is slow, subjective, and provides limited quantitative feedback. This work presents a lightweight, single-camera computer vision system for automated volleyball serve analysis from standard baseline video. The pipeline detects and tracks the volleyball with a fine-tuned YOLOv8 object detector combined with SORT-based tracking and trajectory post-processing. Serve contact and landing frames are estimated automatically from smoothed ball trajectories using size- and motion-based heuristics, while court geometry is recovered through manual annotation of visible court corners and homography-based calibration. This enables projection of serve trajectories and landing locations into real-court coordinates, supporting accuracy, consistency, and zone-based analysis. Evaluation on a privately collected dataset of annotated serves shows a mAP50–95 of 0.911 for ball detection and landing-frame estimates within approximately one frame of human-labeled ground truth. These results demonstrate that accurate, quantitative volleyball serve analysis is achievable using only a single camera and lightweight computer vision techniques.

Index Terms—Computer vision; Volleyball analysis; Ball tracking; Sports analytics; Machine learning; YOLO; SORT; Trajectory analysis

I. INTRODUCTION

Video analysis has become an integral part of modern sports training, offering athletes and coaches valuable feedback on performance. However, existing tools for volleyball are often limited in precision, accessibility, or scope. In particular, serve analysis remains a largely manual process, relying on subjective visual review rather than quantitative, automated measurements. This project explores how computer vision can be leveraged to develop a lightweight, single-camera system for automated serve analysis, directly from raw video.

A. Volleyball Overview

Volleyball is a fast-paced team sport that relies on precision, timing, coordination, and teamwork. Each rally starts with a serve, followed by a sequence of passes, sets, and attacks that aim to ground the ball on the opponent’s court. A standard indoor court measures 18 by 9 meters, with a net that is 2.43 meters high in men’s play and 2.24 meters high in women’s play. Teams rotate through six positions, with each player alternating between serving, receiving, and front-row attacking roles throughout a match. Thus each player must master their serve for the team to succeed.

Despite its simple appearance, volleyball is defined by highly technical movements. Serves, spikes, and passes all

depend on millisecond-level timing and consistent mechanics. This makes volleyball an interesting challenge for computer vision: tracking a small, fast-moving ball through rapid changes in direction, occlusions, varied lighting, and motion blur while also understanding spatial context on the court. The presence of players moving across the frame, partially obscuring the ball, adds further complexity to automated analysis.

B. Importance of the Serve

The serve is the only action in volleyball that is completely under one player’s control, making it one of the most critical components of the game. Each player on the court is guaranteed to serve at least once per set, and serves can directly score points if they land on the opponent’s court. A strong serve can pressure the opponent’s receive formation, while a weak or inconsistent one often results in free points for the opponent. Because every rally begins with a serve, it often dictates early momentum and sets the tone for subsequent plays. Small variations in toss height, contact angle, or approach can shift the landing zone by several meters. Because these differences occur over tens of milliseconds, athletes rely heavily on video to refine technique. Yet traditional video review provides only qualitative feedback such as “a little short” or “drifted wide,” offering no quantitative measure of accuracy or consistency across an entire practice session.

C. Motivation for Automated Analysis

Some commercial tools, such as BallTimeAI [1], attempt to automate volleyball analysis, but they are closed-source, subscription-based, and focused mainly on match statistics rather than detailed serve mechanics. Many also rely on specialized equipment or multi-camera setups, which limits accessibility for teams that record practices using a single baseline camera.

Computer vision provides a practical alternative. With accurate ball tracking and court calibration, a system can extract useful serve information such as trajectory shape, toss consistency, and landing accuracy from ordinary video. The main difficulty is that the ball is small, fast, and frequently blurred or occluded, which makes detection unreliable from a single viewpoint. Robust tracking must handle missing or noisy observations while still maintaining a stable trajectory estimate.

To address this, the system uses a fine-tuned YOLOv8 detector [2]. Training on full-resolution 1080p footage allows the model to handle blur, partial occlusion, and rapid motion, achieving a mAP50–95 of 0.911 with over 99% precision and recall.

D. Project Scope and Objectives

This project develops a single-camera system that processes baseline volleyball footage and automatically extracts serve metrics. The final system delivers four core capabilities:

- **Ball Detection and Tracking:** A fine-tuned YOLOv8 model integrated with SORT [3] and smoothing to recover stable trajectories.
- **Hit and Landing Estimation:** A size–motion heuristic that identifies the moment of contact and the landing frame, achieving accuracy within approximately ± 1 frame.
- **Court Calibration:** Homography computed from manually annotated court corners to convert pixel measurements into real-court coordinates.
- **Serve Accuracy and Consistency Analysis:** Landing-point visualizations, trajectory overlays, and zone-based summaries for players and sessions.

Additional components such as automatic serve isolation and court-line segmentation were explored conceptually but deferred to future work.

Across 113 annotated serves collected from six players and five recording sessions, the system produces stable trajectories and landing estimates within approximately ± 1 frame of human-labeled ground truth. These experiments demonstrate that accurate, quantitative serve analysis is achievable using only a baseline camera and lightweight computer vision methods.

Together, these components form a functional baseline system for automated serve evaluation and demonstrate the feasibility of single-camera volleyball analytics. The system provides objective and repeatable feedback for training and creates a foundation for future extensions.

II. RELATED WORK

Computer vision has been widely applied across sports analytics to extract performance metrics, identify tactical patterns, and assist coaching. Many systems rely on object detection models such as YOLO, which are effective at identifying small, fast-moving objects under varying lighting and camera conditions. YOLO’s single-shot architecture enables real-time inference and has been adopted in numerous sports-tracking applications. Temporal association methods such as SORT [3] complement such detectors by linking frame-by-frame predictions into coherent trajectories using simple motion models. Together, detection and tracking form the basis for many sports ball-tracking pipelines.

In tennis, TrackNet [4] introduced a heatmap-based ball localization approach that predicts per-pixel probability maps. This architecture enables robust tracking under strong motion blur and rapid directional changes, outperforming traditional

bounding-box detection for small-object motion. However, TrackNet is tailored to tennis broadcasts, where ball scale is larger, background clutter is lower, and camera perspectives are standardized. Volleyball practice footage presents different challenges: players frequently occlude the ball, camera positions vary significantly, and the ball may occupy only a few pixels. These factors limit the direct applicability of TrackNet to detailed serve analysis.

Volleyball-specific computer vision research has expanded in recent years. Plankelj and Mlakar [5] proposed a system using U-Net for court segmentation and YOLOv8 for ball detection, combined with homography to create tactical visualizations. Their results show that convolutional models can detect court geometry and ball trajectories even when image quality varies. However, their system focuses on match-level ball movement rather than serve mechanics, and it assumes visible court lines for segmentation, which is not guaranteed in typical baseline practice recordings.

More advanced volleyball research explores monocular 3D trajectory reconstruction. Dong et al. [6] and similar works fuse temporal features with physics-based constraints to infer ball depth and full 3D trajectories. These approaches support applications such as set-type recognition or action prediction, but typically require high-quality video, predictable motion patterns, or multiple visible bounces. Their modeling complexity makes them less suitable for training environments where teams rely on handheld or smartphone recordings.

Commercial systems such as BallTimeAI [1] also provide automated volleyball analysis using proprietary models. These systems detect players and the ball, classify actions, and generate match summaries. However, they emphasize match analytics rather than technical skill evaluation. Because they are closed-source and rely on undisclosed datasets, they cannot be easily extended or validated by researchers, and they do not provide serve-level landing metrics or consistency measurements.

Despite progress in general ball tracking, volleyball match analysis, and monocular trajectory estimation, serve-focused analysis remains largely unexplored. Existing systems either require complex multi-camera setups, address general ball motion rather than serve mechanics, or do not provide quantitative landing metrics. The present work addresses this gap by combining a fine-tuned YOLO detector, lightweight tracking, and court calibration to produce serve-level trajectory and landing estimates from a single baseline camera.

III. DATASET AND ANNOTATION

A. Recording Setup

All data was collected using a fixed single-camera baseline setup designed to capture the server, ball flight, and landing zone in a single view. The camera was positioned at the center of the baseline, approximately 4 m behind the court, mounted on a 74 in (188 cm) tripod and tilted downward by roughly 5–10°. This ensured that both the near and far court lines remained visible while minimizing perspective distortion.

Recordings took place at the Rochester Institute of Technology Hale-Andrews Student Life Center volleyball courts under overhead LED lighting. Lighting conditions varied slightly between sessions, but no exposure correction or stabilization was applied; built-in video stabilization was disabled to prevent frame warping during rapid ball motion. All recordings were captured on a Samsung Galaxy S23 at 1920×1080 resolution and 60 FPS, providing sufficient temporal resolution for tracking fast serves. Across six players and five sessions, a total of 113 serve clips were collected. This consistent viewpoint formed a standardized dataset suitable for training and evaluating the ball detector, tracking system, and landing estimator.

B. Data Organization for Model Development

The primary research dataset is stored in the `data/` directory, which contains raw videos, processed serve clips, extracted frames, annotations, trajectories, and metadata used for training and validating the YOLO detector and testing the full pipeline. Its structure is shown in Fig. 1.

Raw video recordings are stored in `data/videos/raw/YYYY-MM-DD/session_N/`, preserving original capture dates and ensuring reproducibility. Individual serves are manually segmented using `split_serves.py` and stored under `data/videos/processed/<player>/session_N/` with a consistent naming scheme (`serve_001.mp4`, `serve_002.mp4`, etc.). Approximately 16,000 frames were extracted across all serves and stored in `data/frames/` for frame-level annotation and YOLO training.

Annotation files reside in `data/annotations/`. Ball-detection labels (YOLO format) and model-generated detection JSONs are stored in `ball_detections/`, while `court_corners/` contains per-session calibration JSON files used to compute homographies. Ball trajectories produced by SORT are saved under `data/trajectories/<player>/session_N/`.

Metadata linking players, sessions, and serves is maintained in `data/metadata/`. The `players.csv` file contains basic player information, and `serves.csv` records serve identifiers, video paths, start and end frames, and manually annotated landing frames. This table is automatically updated by utilities such as `split_serves.py` and `landing_frame.py`, ensuring consistent indexing throughout the pipeline. The `datasets/` directory contains the formatted YOLO train/validation splits used during training and hyperparameter tuning.

C. User-Facing Dataset Structure

While `data/` contains the full research dataset, the end-user workflow operates entirely within the `user/` directory. This separation allows a coach or athlete to run serve analysis without interacting with the training data or internal development files.

The `user/` directory mirrors the structure of the research dataset but stores only the user's own processed videos, detections, trajectories, and results. Its main components include:

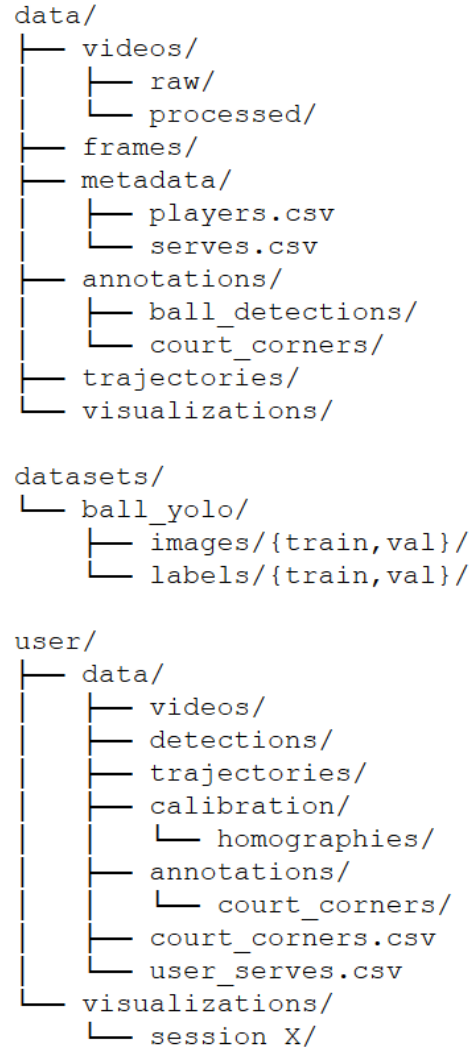


Fig. 1. Directory structure.

- **user/data/videos/**: serve clips split from user-provided raw videos.
- **user/data/annotations/court_corners/**: session-specific court-corner JSON files created interactively using `annotate_court.py`.
- **user/data/detections/** and **user/data/trajectories/**: YOLO detection results and SORT tracking outputs.
- **user/data/calibration/homographies/**: homography files mapping image coordinates to real-court coordinates.
- **user/data/user_serves.csv**: a lightweight metadata table tracking serve IDs, hit and landing frames, and landing coordinates.
- **user/visualizations/**: output images such as landing heatmaps, trajectory overlays, and zone-accuracy visualizations.

The automated `workflow.py` script reads and writes exclusively within the `user/` subtree, ensuring that model development and end-user analysis remain clearly separated.

D. Annotation Process

Ball detection labels were created using the open-source Computer Vision Annotation Tool (CVAT) [7]. Annotators drew bounding boxes around the volleyball on selected frames, using CVAT's interpolation and tracking features to propagate annotations across sequences. Labels were exported in YOLO format and integrated directly into the training pipeline.

As the detector improved, intermediate YOLO models were used for semi-automatic annotation. The `auto_label_and_upload.py` script generated predicted bounding boxes for new frames, packaged them into CVAT tasks, and allowed annotators to correct only the remaining errors.

Temporal landing labels were collected using the `landing_frame.py` tool, which records the first contact frame for each serve. These labels provide ground truth for evaluating the landing-frame estimator.

Court geometry was annotated once per session using `annotate_court.py`. Annotators selected four court corners and two center-line points, which were saved to JSON files in `court_corners/`. Homographies computed from these annotations allow all trajectories and landing points to be projected into physical court coordinates (meters).

Together, the recording setup, organized dataset structure, and multi-stage annotation pipeline yield a unified dataset supporting all detection, tracking, and serve-analysis experiments in this project.

IV. METHODOLOGY

This section describes the processing pipeline used to detect the volleyball, track its motion, calibrate image coordinates to real court coordinates, and estimate hit and landing events for each serve. The pipeline follows a modular design that supports debugging and future extensions, and consists of five stages: ball detection using a fine-tuned YOLOv8 model, temporal tracking with SORT and custom fragment merging, court calibration via homography estimation, trajectory smoothing and event detection, and projection into real-world coordinates for visualization and analysis.

A. Ball Detection (YOLOv8)

Ball detection is performed using a YOLOv8n model fine-tuned specifically for volleyball serve footage. YOLOv8 provides a lightweight single-stage detector capable of operating at full 1080p resolution in real time, making it suitable for small, fast objects such as volleyballs that may appear blurred or partially occluded. The model was fine-tuned on 16,172 frames extracted from 113 serves across six players. Frames were annotated with YOLO-format bounding boxes using CVAT, including both manual labels and model-assisted annotations from intermediate checkpoints.

Training was conducted using the Ultralytics framework with a batch size of 4 and 100 epochs, retaining the pre-trained COCO backbone and optimizing only the detection heads. During inference, frames are processed at the original 1920×1080 resolution. Detections below a confidence of 0.3

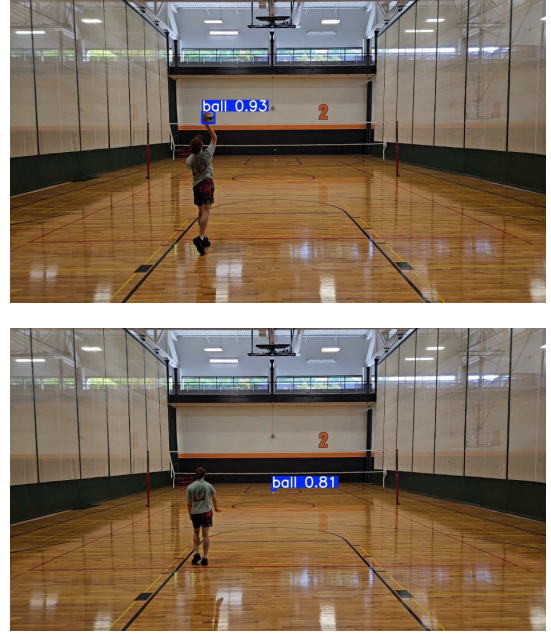


Fig. 2. Example YOLOv8 detections on 1080p frames. The ball is often small, blurred, or partially occluded. (a) Ball detection on serve, (b) Ball detection on landing

or failing NMS at IoU 0.65 are discarded. Each retained detection is stored in JSON with center coordinates, bounding-box size, and confidence.

Figure 2 shows sample detections on frames with occlusions and blurred, miniscule (less than 10 pixels) balls, illustrating the robustness of the fine-tuned detector.

B. Ball Tracking with SORT and Fragment Merging

Although YOLOv8 provides accurate per-frame detections, temporal linking is necessary to form coherent ball trajectories. We use the SORT tracker, which applies a Kalman filter to predict object motion and uses IOU-based Hungarian assignment for frame-to-frame consistency. SORT is well-suited for this task: it is fast, has no re-identification component (unnecessary for a single-object scene), and naturally handles moderate motion blur.

SORT is configured with `max_age=30` (allowing up to 30 missed detections before terminating a track), `min_hits=1`, and `iou_threshold=0.3`. The output of SORT is typically a set of short, fragmented tracks. These fragments arise primarily from occlusions by the server's arm or torso, failures of the detector during the ball's fastest motion, or abrupt appearance changes as the ball rotates.

To reconstruct a single continuous trajectory, we introduce a greedy fragment-merging procedure. Two fragments are considered merge candidates if their temporal gap falls within `[-overlap_back, +frame_gap]` and the Euclidean distance between the earlier fragment's endpoint and the later fragment's start point is below a spatial threshold. A cost function combining temporal gap and spatial distance ranks merge candidates. Edges are accepted greedily while enforcing a

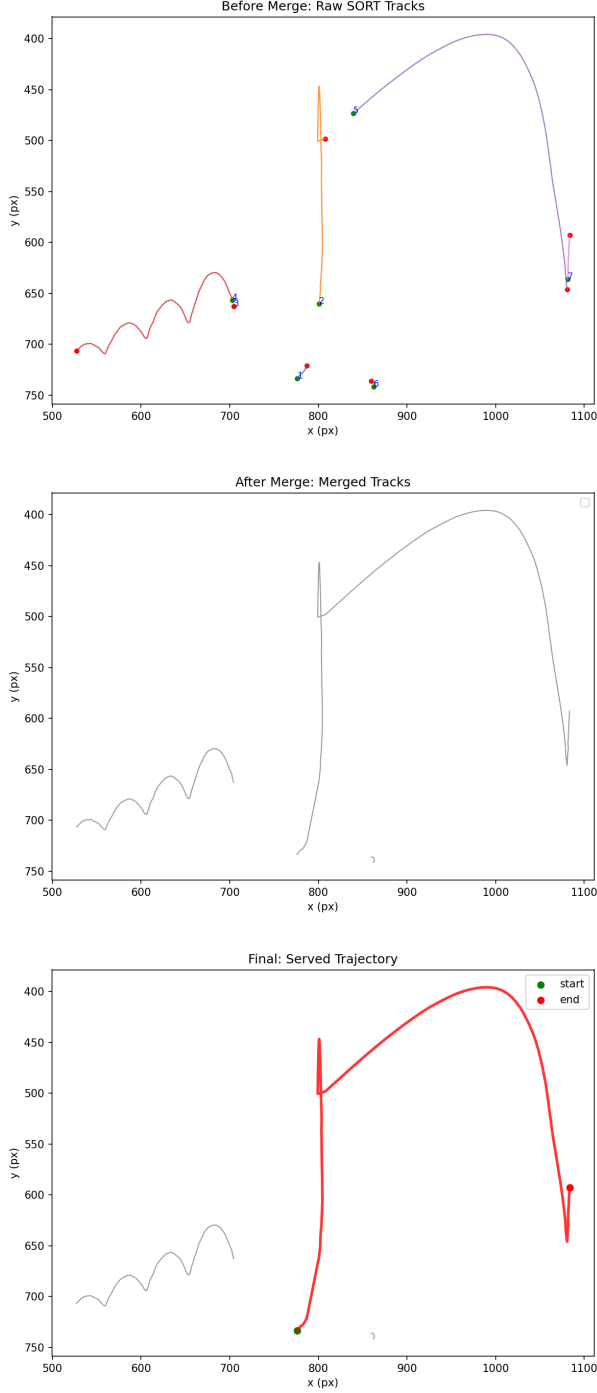


Fig. 3. Trajectory reconstruction stages from detection to final serve path. (a) Raw SORT fragments, (b) Merged trajectories, (c) Selected serve trajectory

one-incoming, one-outgoing constraint for each fragment and preventing cycles.

Among the merged trajectories, the final serve path is selected using a simple speed-based heuristic that favors longer, faster trajectories. It also ensures the average size of the ball decays through the track, preventing a rolling ball becoming the selected track. Figure 3 visualizes the three stages of this

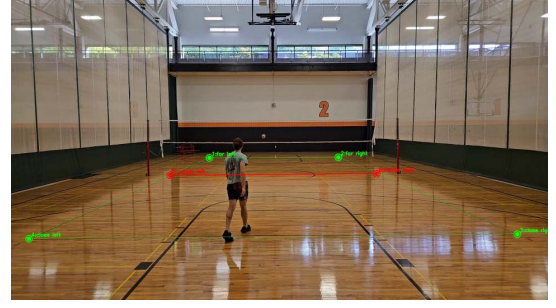


Fig. 4. Manual annotation of six court reference points allows pixel-to-meter calibration via homography.

process: raw SORT fragments, merged sequences, and the final selected trajectory.

C. Court Calibration via Homography

To convert image-space trajectories into physical court coordinates, each recording session is calibrated using a planar homography. Unlike methods that require court-line segmentation or multi-camera geometry, our approach uses a simple manual annotation step: the user clicks six distinctive court points visible in the frame using an interactive tool. These points include the four court corners and the two center-line intersections. Their corresponding world coordinates are defined using standard volleyball dimensions, with the origin at the near-left baseline corner, the x -axis spanning the nine-meter width, and the y -axis extending the full eighteen-meter court length, as shown in Figure 4.

Given the image–world correspondences, a 3×3 homography matrix H is computed and stored for all serves in that session. Applying H transforms any pixel coordinate (u, v) into metric coordinates (x, y) , enabling realistic trajectory overlays, landing-zone assignment, and metric consistency across sessions.

D. Trajectory Smoothing

The selected trajectory consists of per-frame measurements of the ball’s center coordinates and apparent size (computed as $\sqrt{wh}$). These sequences contain noise from detection jitter, bounding-box variation, and tracking uncertainty. To obtain smooth and differentiable signals, we apply a Savitzky–Golay filter [8] with an adaptive odd window length up to 9 and polynomial order 2. This produces smoothed horizontal and vertical positions, $x_s(t)$ and $y_s(t)$, and smoothed apparent size $s(t)$.

The derivative of vertical position, dy/dt , is computed using a finite-difference approximation. This signal is particularly important for landing detection, as a shift from downward to upward velocity after a hit often indicates ground contact and rebound.

E. Hit and Landing Analysis

The final stage of the pipeline identifies two critical temporal events: the *hit frame*, corresponding to the player’s contact with the ball, and the *landing frame*, corresponding to the

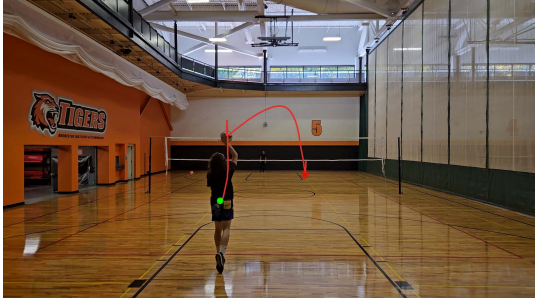


Fig. 5. Example ball trajectory projected to real-world coordinates using session-specific homography.

ball’s first contact with the ground. Both are inferred from the smoothed trajectory signals and their derivatives.

1) *Hit Detection*: Hit detection uses a hybrid motion–appearance heuristic. Frames before index 20 are ignored to remove early toss noise. The algorithm then identifies a “near-server” region where the apparent size exceeds 90% of its maximum; in this region the ball is physically closer to the camera, encompassing both the toss apex and the hit moment.

Within this region, the toss apex is the minimum of $y_s(t)$ and apparent size. After the apex, the algorithm searches for the earliest window in which apparent size decreases consistently and significantly, indicating that the ball has been struck and is accelerating away from the camera.

This approach is robust to partial occlusions and brief detection dropout, since size changes often persist even when positional noise is present.

2) *Landing Detection*: Landing detection is performed within a temporal window following the estimated hit frame. Candidate landing frames are identified primarily using the vertical position signal, where the first prominent local maximum in $y(t)$ corresponds to ground contact and rebound. To reduce false positives caused by mid-flight noise or tracking instability, candidates are additionally constrained by apparent ball size: only frames where the smoothed size is below a fixed fraction (0.5) of the maximum observed size are considered valid. This ensures that landing detection occurs only when the ball is sufficiently far from the camera, consistent with floor contact on the far side of the court. As observed in Figure 6 the system combines the use of $y_s(t)$, dy/dt , and apparent size in making these calculations.

If no valid vertical-position peak satisfies this size constraint, a fallback strategy is used based on a rolling minimum of the apparent ball size. The candidate frame is refined by selecting the local frame with the largest vertical position while maintaining the size constraint. A final sanity check enforces that the landing frame occurs strictly after the hit frame, preventing degenerate estimates in truncated or failed tracks.

F. Visualization and Zone Analysis

Once hit and landing frames are identified, the landing pixel coordinate is projected through the session’s homography to

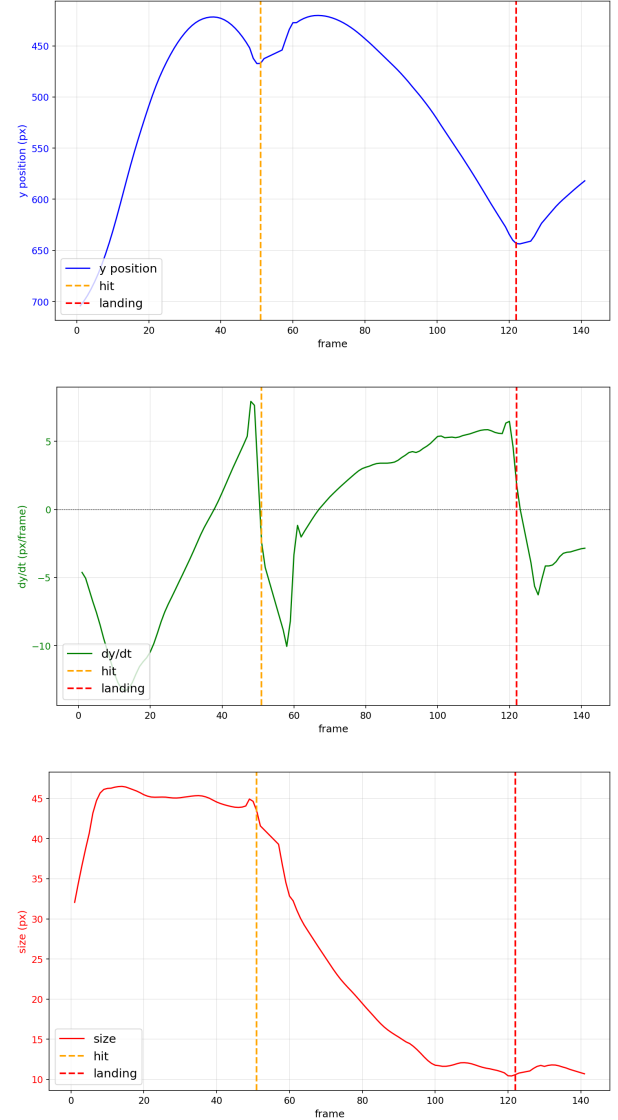


Fig. 6. Smoothed vertical position, derivative, and apparent size signals used to detect hit and landing events. (a) Vertical position, (b) Derivative, (c) Apparent size

obtain metric (x, y) coordinates on the far side of the court. These coordinates enable serve-level analytics, including trajectory plots (Figure 5), landing-point scatter maps, density heatmaps, and accuracy evaluation within FIVB-style zones. All visualizations are produced automatically as part of the user workflow.

The combination of detection, tracking, calibration, and temporal analysis yields a fully automated system capable of extracting quantitative serve characteristics from a single baseline camera.

V. RESULTS AND EVALUATION

A. Dataset Summary and Evaluation Setup

The system was evaluated on a private dataset consisting of 113 annotated serves recorded across six players and five

TABLE I
YOLOv8 BALL DETECTION PERFORMANCE (BEST EPOCH @ 90)

Metric	Value
Precision	0.9452
Recall	0.99527
mAP50	0.99495
mAP50–95	0.91115

recording sessions. All videos were captured using a fixed baseline camera at 1920×1080 resolution and 60 FPS. For model training and evaluation, the dataset was split into 80% training and 20% validation sets at the serve level. Evaluation focuses on the accuracy and stability of each stage of the pipeline, including ball detection performance, trajectory reconstruction quality, landing frame estimation accuracy, court calibration reliability, and serve-level consistency analysis. Because no public single-camera volleyball serve dataset with comparable annotations was found, results are reported relative to human-labeled ground truth and qualitative inspection rather than external benchmarks.

B. Ball Detection Performance

Ball detection performance was evaluated using standard object detection metrics on a held-out validation set. The fine-tuned YOLOv8 detector achieved a mAP50–95 of 0.911, with precision and recall of 0.9945 and 0.9953 on the validation set, as seen in Table I. These results indicate that the detector reliably identifies the volleyball under challenging conditions.

Figure 2 illustrates representative detection results across several difficult frames. The examples highlight cases where the ball occupies fewer than ten pixels or overlaps with players. Despite these challenges, the detector consistently localizes the ball with high confidence, providing stable inputs for downstream tracking and trajectory analysis.

C. Tracking and Trajectory Quality

Temporal tracking is performed using SORT, which associates detections across frames using motion prediction and spatial overlap. In practice, SORT frequently produces fragmented tracks when detections are briefly lost due to occlusion, motion blur, or rapid changes in ball velocity following contact events. To address this, a custom fragment-merging strategy is applied that reconnects track segments based on temporal proximity, spatial continuity, and consistency in motion.

Figure 3 visualizes this process in three stages: the raw SORT track fragments, the merged trajectories after post-processing, and the final selected serve trajectory. After merging, the system reliably recovers a single continuous trajectory corresponding to the served ball. A speed-weighted and length-aware selection criterion is then applied to isolate the serve trajectory from remaining spurious tracks, such as those caused by stray balls or brief false detections.

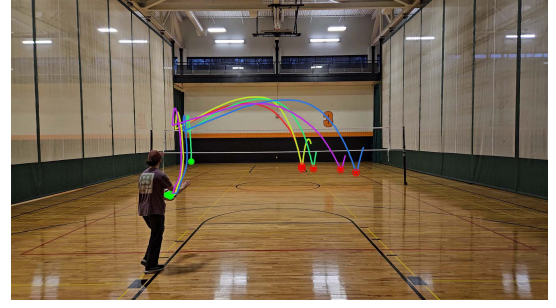


Fig. 7. Example overlay of projected trajectories and landing points. Each trajectory is color-coded by serve ID.

D. Hit and Landing Estimation Accuracy

Landing frames were manually annotated for all serves and used as ground truth for evaluation. Hit frames were estimated automatically and were not included in the manual annotation process. Landing estimation accuracy is measured as the absolute frame difference between the predicted landing frame and the annotated ground truth.

Figure 6 shows representative signal plots of vertical position, vertical velocity, and apparent ball size over time for a single serve. The predicted hit and landing events are overlaid on these signals, illustrating how motion-based cues and size constraints jointly contribute to robust landing detection. Across the dataset, predicted landing frames fall within approximately one frame of the annotated ground truth on average, with the majority of serves falling within a two-frame error window.

E. Court Calibration Accuracy

Court calibration is performed once per recording session using manually annotated court corners and center-line points. The resulting homography maps image coordinates to real-world court coordinates measured in meters. Calibration accuracy is assessed qualitatively by projecting trajectories and landing points onto a rendered court and verifying alignment with visible court lines.

F. Serve Accuracy and Consistency Analysis

Using calibrated landing coordinates, the system generates serve-level accuracy and consistency visualizations for each player and session. Figure 7 shows an example frame with all serve trajectories from a single session overlaid, with estimated hit and landing points marked for each serve. This visualization provides an intuitive overview of serve repeatability, directional bias, and variation across attempts.

In addition, landing distributions are summarized using court-projected heatmaps. Figure 8 highlights spatial clustering patterns on the far side of the court, revealing both targeting tendencies and serve consistency across a session. To further contextualize these distributions, Figure 9 visualizes the same projected landing locations separated by predefined court zones. Serves landing in a user-specified target zone are highlighted in green, while all other serves are shown in red.

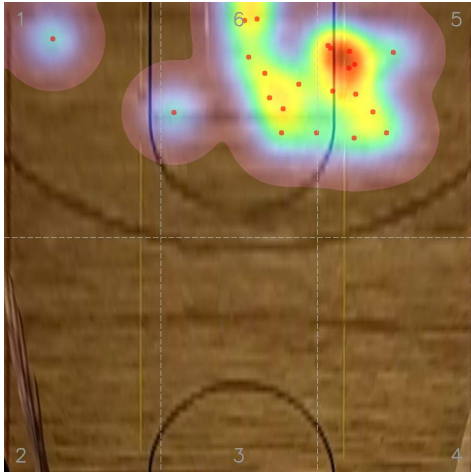


Fig. 8. Court-projected landing heatmap for a single session, with color intensity indicating spatial density.

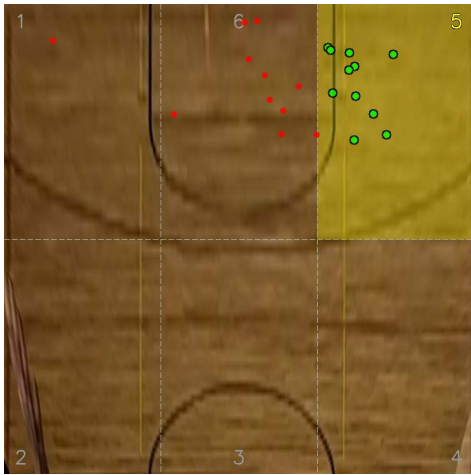


Fig. 9. Zone-based landing analysis for a single session.

This representation directly supports quantitative zone-based analysis, enabling summaries such as target-zone percentage and per-serve landing coordinates to be computed automatically. Together, these visualizations demonstrate that meaningful serve-level insights, including accuracy, consistency, and targeting behavior, can be extracted using only a single baseline camera and lightweight computer vision techniques.

G. Failure Cases and Limitations

Failure cases primarily occur when the ball is fully occluded for extended periods, when detection confidence drops for several consecutive frames, or when the ball contacts the net and sharply alters its trajectory. In these cases, tracking fragmentation or ambiguous landing signals can lead to missed or inaccurate estimates. Despite these limitations, the system remains robust across the majority of serves recorded under realistic practice conditions.

VI. CONCLUSIONS

A. Conclusions

This project demonstrates that accurate volleyball serve analysis can be achieved using a single baseline camera and lightweight computer vision techniques. By combining a fine-tuned YOLOv8 detector, robust multi-object tracking with trajectory merging, and homography-based court calibration, the system reliably extracts serve-level metrics directly from standard practice footage.

Across 113 annotated serves from six players and five recording sessions, the pipeline produces stable ball trajectories and estimates landing frames within approximately one frame of human-labeled ground truth. Court-projected landing locations enable quantitative analysis of serve accuracy, spatial consistency, and zone-based targeting, providing insights that are difficult to obtain through manual video review alone.

Importantly, the system requires no specialized hardware, multi-camera setups, or external sensors. All processing operates on 1080p video recorded from a single fixed viewpoint, making the approach accessible to individual athletes, coaches, and smaller programs. These results demonstrate the feasibility of practical, single-camera volleyball analytics and establish a functional baseline for future extensions in sports performance analysis.

B. Future Work

Several directions exist to extend and improve the current system. First, manual serve splitting and court annotation remain the primary sources of user interaction. Fully automatic serve isolation using pose estimation or motion cues, as well as automatic court line detection, would further reduce setup time and improve usability.

Second, while the current system estimates 2D trajectories and landing locations, the single-camera setup limits direct depth recovery. Future work could explore approximate 3D trajectory reconstruction by calibrating ball size across depth at the start of a session or by incorporating simple physical flight models to infer height and velocity.

Robustness under challenging conditions also remains an open area. Extended occlusions, severe motion blur, net interactions, and abrupt trajectory changes can still degrade tracking quality. Integrating temporal learning-based trackers or physics-aware constraints may help address these failure cases.

Finally, real-time or near-real-time feedback represents a promising application. With further optimization, the pipeline could provide immediate serve accuracy statistics and visualizations during practice, enabling athletes to adjust technique within a session rather than during later review.

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